

Test Report - Leaf Log

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1.0 Introduction

1.1 Purpose

The purpose of this test report is to evaluate the functionality, usability, and user interface of the Leaf Log Application. This report aims to carry out the first round of tests that could help identify and document issues, bugs, or inconsistencies within the application. Through functionality and interface testing, we can determine whether the app will work as expected.

2.0 Test Summary

Leaf Log meets all the requirements for the project consisting of at least 4 remote CRUD APIs, 4 local APIs, proper navigation and sensor-related functionalities.

Proper Credentials for Testing:

- **Name:** Test User
- **Password:** 12345#Password
- **Journal Title:** Journal
- **Journal Description:** This is the primary journal used for testing.
- **Log Title:** Log
- **Log Description:** This is the primary log used for testing.

Authentication System

- The authentication system serves as a password system only for the device it is on. Accounts registered do not have any content associated with them. For instance, using the account registered on Device A can be logged into using Device B but journals made on Device A cannot be accessed in Device B.
- Leaf Log's system follows the expected authentication flow and mostly works as expected. Minor improvements in user support could be made to enhance user experience.
- Leaf Log lacks an onboarding system so there are functionalities that are not as intuitive despite its simplicity.
- Leaf Log does not cache user credentials thus requiring users to constantly log in every time they open the app which may be inconvenient in the long run.

Functionalities

- The application generally works as expected with the majority of its functionalities working to record their plant information. Despite being simple and easy to use after a while of usage, there are still improvements to be made to make it more intuitive for new users due to the ambiguity of some functionalities.
- A caching system is present where actions done without an internet connection are synced properly once it regains the internet connection.
- More user support could be done to account for unexpected events.

Design Considerations

- Design issues pose more issues on the emulator than on a physical device due to minor dynamic issues such as the top navigation or the colour scheme. Because of these, some tests could only be conducted on a physical device as it was simply not working on the emulator.
- Two different themes were implemented in the application to emphasise the feeling of a journal which is unique but the constant switching between different styles may confuse new users as well as the implementation of higher contrast colours.
- Nice touch of animation to transition between pages.
- Locks screen to prevent landscape orientation errors.

2.1 Authentication System

2.1.1 Test Case: Functional Testing - Registration Screen

The main purpose of this test case is to check if the registration screen performs its expected function of registering new users into the application and to check for form validity and user support. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Register an account using the proper credentials
- b. Register another account using the credentials used previously
- c. Register another account using the same name as above but a different password
- d. Register an account with passwords that don't match
- e. Register an account with a bad password
- f. Register an account without filling in anything
- g. Register an account without filling in the name
- h. Register an account without filling in the password
- i. Register an account using different emojis as the name
- j. Testing other buttons present on the screen

Test Results:

- a. Register an account using the proper credentials
 - A circular loading indicator appears on top of the text fields
 - The registration screen routes to the journal screen
- b. Register another account using the credentials used previously
 - A snack bar appears saying that the user is already registered
- c. Register another account using the same name as above but a different password
 - A snack bar appears saying that the user is already registered
- d. Register an account with passwords that don't match
 - The supporting text appears saying that both passwords do not match
- e. Register an account with a bad password
 - "password" - Produces an internal server error snack bar
 - "123456" - Produces an internal server error snack bar
- f. Register an account without filling in anything
 - The supporting text appears saying that the field is required
- g. Register an account without filling in the name
 - The supporting text appears saying that the field is required
- h. Register an account without filling in the password
 - The supporting text appears saying that the field is required
- i. Register an account using different emojis as the name
- j. Testing other buttons present on the screen
 - Successfully routes back to Login and Offline Mode
 - Hide/View Password works

Additional Comments: Registration form validation and user support are generally well done but a more comprehensive message is needed to inform users of which bad passwords are not accepted.

2.1.2 Test Case: Interface Testing - Registration Screen

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Analyse Contrast
- b. Analyse Repetition
- c. Analyse Alignment
- d. Analyse Proximity
- e. Analyse Aesthetic
- f. Analyse Intuitiveness

Test Results:

- a. Analyse Contrast
 - Colour contrast is present
 - Primary buttons are distinguishable from secondary buttons
- b. Analyse Repetition
 - Consistent formatting of elements
- c. Analyse Alignment
 - The screen follows a centre alignment
- d. Analyse Proximity
 - Elements are spaced out evenly
- e. Analyse Aesthetic
 - The screen follows the material 3 green theme
 - Simple UI
- f. Analyse Intuitiveness
 - Follows the standard convention of a registration system
 - The arrow icon in the register buttons feels out of place, it could give the impression that by clicking it the user will route TO the register screen instead of registering an account

Additional Comments: Registration screen design is generally well done but more experimentation can be done to try out other styles of elements due to basic interface. Odd placement of circular indicators.

2.1.3 Test Case: Functional Testing - Login Screen

The main purpose of this test case is to check if the login screen performs its expected function of logging in only existing users into the application and to check for form validity and user support. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Login to an account using the proper credentials previously registered
- b. Login to an account using the proper credentials previously registered but with an incorrect password
- c. Login to an account that does not exist
- d. Closing the phone and opening the application
- e. Testing other buttons present on the screen

Test Results:

- a. Login to an account using the proper credentials previously registered
 - Successfully routes to the Journals Screen
- b. Login to an account using the proper credentials previously registered but with an incorrect password
 - Produces an internal server error snack bar
- c. Login to an account that does not exist
 - Produces an internal server error snack bar
- f. Closing the phone and opening the application
 - No login cache is present so every time the user opens the application they have to log in which can be inconvenient/frustrating
- d. Testing other buttons present on the screen
 - Successfully routes back to Login and Offline Mode
 - Hide/View Password works

Additional Comments: Login form validation is generally well done but there is little user support for unsuccessful login attempts. A more comprehensive message is needed to inform users of what errors are occurring.

2.1.4 Test Case: Interface Testing - Login Screen

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Analyse Contrast**
- b. Analyse Repetition**
- c. Analyse Alignment**
- d. Analyse Proximity**
- e. Analyse Aesthetic**
- f. Analyse Intuitiveness**

Test Results:

- a. Analyse Contrast**
 - Colour contrast is present
 - Primary buttons are distinguishable from secondary buttons
- b. Analyse Repetition**
 - Consistent formatting of elements
- c. Analyse Alignment**
 - The screen follows a centre alignment
- d. Analyse Proximity**
 - Elements are spaced out evenly
- e. Analyse Aesthetic**
 - The screen follows the material 3 green theme
 - Simple UI
- f. Analyse Intuitiveness**
 - Follows the standard convention of a login system
 - The arrow icon in the login buttons feels out of place, it could give the impression that by clicking it the user will route TO the login screen instead of logging into an account

Additional Comments: Login screen design is generally well done but more experimentation can be done to try out other styles of elements due to the basic interface.

2.2 Journals System

2.2.1 Test Case: Functionality Testing - Journal Creation (FAB)

The main purpose of this test case is to check if the floating action button performs its expected function of creating new journals. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Create a new journal by filling in the input fields properly**
- b. **Create a new journal but without an image**
- c. **Create a new journal but without a title**
- d. **Create a new journal but without a description**
- e. **Create a new journal but with abnormal title inputs**
- f. **Create a new journal but with abnormal description inputs**
- g. **Testing other buttons present on the screen**

Test Results:

- a. **Create a new journal by filling in the input fields properly**
 - Successfully routes back to the Journal screen, displaying the newly made journal
- b. **Create a new journal but without an image**
 - The supporting text appears saying that the field is required
- c. **Create a new journal but without a title**
 - The supporting text appears saying that the field is required
- d. **Create a new journal but without a description**
 - The supporting text appears saying that the field is required
- e. **Create a new journal but with abnormal title inputs**
 - **Emojis:** Successfully routes back to the Journal screen, displaying the newly made journal with the emojis
 - **Super long text:** Successfully routes back to the Journal screen, displaying the newly made journal with the first few characters
 - Field input has a single-line constraint
- f. **Create a new journal but with abnormal description inputs**
 - **Emojis:** Successfully routes back to the Journal screen, displaying the newly made journal with the emojis
 - **Super long text:** Successfully routes back to the Journal screen, displaying the newly made journal with the first few characters. However, there are no box constraints so the text extends infinitely, with the confirm button remaining static thus creating an element overlap.
- g. **Testing other buttons present on the screen**
 - The camera functions well, with the photo taken displayed in the form
 - Unable to access the back button (On emulator)

Additional Comments: Add journal form validation and user support are generally well done but better handling of elements to prevent overlappings could be done. Validation could be done to prevent the creation of duplicate journals. An additional feature to consider would be the ability to access the device's photo library as a camera alternative.

2.2.2 Test Case: Interface Testing - Journal Creation (FAB)

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Analyse Contrast**
- b. **Analyse Repetition**
- c. **Analyse Alignment**
- d. **Analyse Proximity**
- e. **Analyse Aesthetic**
- f. **Analyse Intuitiveness**

Test Results:

- a. **Analyse Contrast**
 - Odd switch in the theme. Red is not usually recommended as a theme or background due to it being heavy on the eyes, distracting, and is commonly associated with errors. There are also potential accessibility issues for those with eye issues.
- b. **Analyse Repetition**
 - Elements themselves are consistent, following the basic element fields. The switch in the red theme was inconsistent as the button was green while everything else was different shades of red.
- c. **Analyse Alignment**
 - The screen follows a centre alignment
- d. **Analyse Proximity**
 - Elements are spaced out evenly
- e. **Analyse Aesthetic**
 - The switch from green to red seems a bit unusual
- f. **Analyse Intuitiveness**
 - When I first clicked on the FAB and the red screen appeared I thought a massive error appeared on the screen. This could happen to other users.

Additional Comments: Journal creation screen formatting is generally well done but colour choices are a bit questionable.

2.2.3 Test Case: Functionality Testing - Journal Screen

The main purpose of this test case is to check if the journals screen performs its expected function of displaying the user's journals. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Create 10 journals to see screen adaptability**
- b. Testing other buttons present on the screen**

Test Results:

- a. Create 10 journals to see screen adaptability**
 - The screen reacts dynamically by scrolling once the elements exceed the view
- b. Testing other buttons present on the screen**
 - Floating action button floats nicely above the elements
 - Journal buttons can be tapped multiple times mid-transition, causing multiple of the same screens to appear. Attempts to navigate back will cause the application to crash.

Additional Comments: Performs as expected, except for when it made the application crash.

2.2.4 Test Case: Interface Testing - Journal Screen

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- g. **Analyse Contrast**
- h. **Analyse Repetition**
- i. **Analyse Alignment**
- j. **Analyse Proximity**
- k. **Analyse Aesthetic**
- l. **Analyse Intuitiveness**

Test Results:

- g. **Analyse Contrast**
 - Colour contrast is present
 - Good choice of colour switch to white on dark red instead of black
- h. **Analyse Repetition**
 - Consistent formatting of elements
- i. **Analyse Alignment**
 - Consistent layout of elements
- j. **Analyse Proximity**
 - Elements are spaced out evenly
- k. **Analyse Aesthetic**
 - Likewise, a bit strange to use error-coloured red journals when everything else in the application is different shades of green
- l. **Analyse Intuitiveness**
 - Journal Icon is intuitive

Additional Comments: Performs as expected.

2.2.5 Test Case: Functionality Testing - Individual Journal Overview

The main purpose of this test case is to check if each journal performs its expected function of displaying the journal information. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Base Journal: Test Display**
- b. **Base Journal: Edit Form Inputs**
- c. **Base Journal: Delete Journal**
- d. **Abnormal Journal: Observing any discrepancies**
- e. **Testing other buttons present on the screen**

Test Results:

- a. **Base Journal: Test Display**
 - Image, title, and description displays as expected
- b. **Base Journal: Edit Form Inputs**
 - Form validation is present for all inputs to ensure everything is filled
- c. **Base Journal: Delete Journal**
 - Successful deletion of journal but no delete confirmation so users may accidentally delete their journals rather easily.
- d. **Abnormal Journal: Observing any discrepancies**
 - For the journal created earlier with really long descriptions and titles, the viewing is fixed in their dimensions so it's inconvenient to keep scrolling such a tiny area of the screen. Users will be unaware of descriptions having multiple paragraphs due to the small restricted view.
- e. **Testing other buttons present on the screen**
 - Unable to access the back button (On emulator)
 - Successfully routes to the Entries Screen.

Additional Comments: Edit journal form validation and user support are generally as expected but better handling of elements could be done to enhance user experience such as prevention of accidental deletions or a chance to recover deleted journals.

2.2.6 Test Case: Interface Testing - Individual Journal Overview

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- m. Analyse Contrast**
- n. Analyse Repetition**
- o. Analyse Alignment**
- p. Analyse Proximity**
- q. Analyse Aesthetic**
- r. Analyse Intuitiveness**

Test Results:

- m. Analyse Contrast**
 - Odd switch in the theme. Red is not usually recommended as a theme or background due to it being heavy on the eyes, distracting, and is commonly associated with errors. There are also potential accessibility issues for those with eye issues.
- n. Analyse Repetition**
 - Elements themselves are consistent, following the basic element fields. The switch in the red theme was inconsistent as the button was green while everything else was different shades of red.
- o. Analyse Alignment**
 - The screen follows a centre alignment
- p. Analyse Proximity**
 - Elements are spaced out evenly
- q. Analyse Aesthetic**
 - The switch from green to red seems a bit unusual
- r. Analyse Intuitiveness**
 - The interface keeps switching from analogue to modern style which may seem confusing for first-time users

Additional Comments: Individual journal display is generally well done but there could be improvements to individual element display and selection.

2.3 Logs System

2.3.1 Test Case: Functionality Testing - Creating Log Entries (FAB)

The main purpose of this test case is to check if the floating action performs its expected function of creating new journal entries. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Create a new log by filling in the input fields properly**
- b. **Create a new log but without filling in certain fields**
- c. **Create a new log but with abnormal inputs**
- h. **Create a new log and get sensor data**
- i. **Testing other buttons present on the screen**

Test Results:

- a. **Create a new log by filling in the input fields properly**
 - Successfully adds a new log and reroutes back to the Entries screen
- b. **Create a new log but without filling in certain fields**
 - Form validation is present with supporting text to indicate that all text fields and images must be filled.
- c. **Create a new log but with abnormal inputs**
 - Number fields account for numbers by remaining empty if the user tries to key in non-number input
- d. **Create a new log and get sensor data**
 - Light sensor performs as expected
 - Temperature and humidity sensors are not working on the emulator. The two sensors being activated also cause the entire application to freeze infinitely
- e. **Testing other buttons present on the screen**
 - Unable to access the back button (On emulator)

Additional Comments: Add log form validation and user support performs as expected but further element considerations could be implemented such as switching to a number keyboard where applicable and providing error messages when the application freezes. Sensor functionalities need to be looked over to determine why it isn't working.

2.3.2 Test Case: Interface Testing - Creating Log Entries (FAB)

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Analyse Contrast**
- b. **Analyse Repetition**
- c. **Analyse Alignment**
- d. **Analyse Proximity**
- e. **Analyse Aesthetic**
- f. **Analyse Intuitiveness**

Test Results:

- a. **Analyse Contrast**
 - Field inputs blend too closely to the background so first-time users may not see all the inputs. However, this issue is tackling through warning supporting text if the necessary inputs aren't keyed in.
- b. **Analyse Repetition**
 - Follows the same structure as creating a new journal
- c. **Analyse Alignment**
 - The screen follows a centre alignment
- d. **Analyse Proximity**
 - Elements are spaced out evenly
 - The distribution of the elements for the get data screen is a bit strange, perhaps a left alignment would be more applicable, especially for many elements with varying widths
- e. **Analyse Aesthetic**
 - Buttons don't match theme
- f. **Analyse Intuitiveness**
 - Confusing for first-time users but easy to familiarise functionalities once users use the application for long periods of time

Additional Comments: Creating log entries display is generally well done but there could be improvements to individual element display and selection.

2.3.3 Test Case: Functionality Testing - Entries Screen

The main purpose of this test case is to check if the entries screen performs its expected function of displaying the user's journal entries. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Create 10 journals to see screen adaptability**
- b. **Testing other buttons present on the screen**
- c. **Other Considerations**

Test Results:

- a. **Create 10 journals to see screen adaptability**
 - The screen reacts dynamically by scrolling once the elements exceed the view
- b. **Testing other buttons present on the screen**
 - Floating action button floats nicely above the elements
- c. **Other Considerations**
 - Log entry titles with really long titles will be covered by the image, and element overlaps present
 - Entry buttons can be tapped multiple times mid-transition, causing multiple of the same screens to appear. Attempts to navigate back will cause the application to crash.

Additional Comments: Generally performs as expected but improvements can be done to enhance the user interface and prevent the application from crashing.

2.3.4 Test Case: Interface Testing - Entries Screen

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Analyse Contrast**
- b. **Analyse Repetition**
- c. **Analyse Alignment**
- d. **Analyse Proximity**
- e. **Analyse Aesthetic**
- f. **Analyse Intuitiveness**

Test Results:

- a. **Analyse Contrast**
 - Colour contrast is present
- b. **Analyse Repetition**
 - Consistent formatting of elements
- c. **Analyse Alignment**
 - Consistent layout of elements
- d. **Analyse Proximity**
 - Elements are spaced out evenly
- e. **Analyse Aesthetic**
 - Unique homely and analogue design
- f. **Analyse Intuitiveness**
 - Journal Entry Icon is intuitive

Additional Comments: Performs as expected.

2.3.5 Test Case: Functionality Testing - Individual Entry Overview

The main purpose of this test case is to check if each journal performs its expected function of displaying and storing each entry log as well as posting it when on an internet connection. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Base Log: Test Display**
- b. **Base Log: Edit Form Inputs**
- c. **Base Journal: Delete Logs**
- d. **Base Log: Post Log to Explore**
- e. **Abnormal Journal: Observing any discrepancies**
- f. **Testing other buttons present on the screen**

Test Results:

- a. **Base Log: Test Display**
 - Log details display as expected
- b. **Base Log: Edit Form Inputs**
 - Form validation is present for all inputs to ensure everything is filled
 - The height allows for decimal placing while water does not
- c. **Base Journal: Delete Logs**
 - Successful deletion of logs but no delete confirmation so users may accidentally delete their logs rather easily.
- d. **Base Log: Post Log to Explore**
 - The share icon does not work on emulator so testing could only be done on a physical device
- e. **Abnormal Logs: Observing any discrepancies**
 - For the logs created earlier with really long descriptions and titles, the viewing is fixed in their dimensions so it's inconvenient to keep scrolling such a tiny area of the screen. Users will be unaware of descriptions having multiple paragraphs due to the small restricted view. Likewise for titles that go beyond the screen.
- f. **Testing other buttons present on the screen**
 - Unable to access the back button (On emulator)
 - Unable to access the share entry button (On emulator)

Additional Comments: Edit journal form validation and user support are generally well done but better handling of elements could be done to enhance user experience.

2.3.6 Test Case: Interface Testing - Individual Entry Overview

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Analyse Contrast**
- b. **Analyse Repetition**
- c. **Analyse Alignment**
- d. **Analyse Proximity**
- e. **Analyse Aesthetic**
- f. **Analyse Intuitiveness**

Test Results:

- a. **Analyse Contrast**
 - Odd switch in the theme. Red is not usually recommended as a theme or background due to it being heavy on the eyes, distracting, and is commonly associated with errors. There are also potential accessibility issues for those with eye issues.
- b. **Analyse Repetition**
 - Elements themselves are consistent, following the basic element fields. The switch in the red theme was inconsistent as the button was green while everything else was different shades of red.
- c. **Analyse Alignment**
 - The screen follows a centre alignment
 - Poor placement of the share icon
- d. **Analyse Proximity**
 - Elements are spaced out evenly
- e. **Analyse Aesthetic**
 - The switch from green to red seems a bit unusual
- f. **Analyse Intuitiveness**
 - The interface keeps switching from analogue to modern style which may seem confusing for first-time users

Additional Comments: Individual journal display is generally well done but there could be improvements to individual element display and selection. Share icon was not spotted, this could happen to many users making them unable to utilise all the functionalities the application provides.

2.4 Explore System

2.4.1 Test Case: Functionality Testing - Explore Screen

The main purpose of this test case is to check if the explore screen performs its expected function of displaying public entries. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Click refresh
- b. Display and view individual entries

Test Results:

- a. Click refresh
 - Screen refreshes accordingly
- b. Display and view individual entries
 - Corresponding public journal entry is displayed with their included information

Additional Comments: Performs as expected. Unable to test whether logs I post are displayed as the post log functionality was not working.

2.4.2 Test Case: Interface Testing - Explore Screen

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- g. Analyse Contrast
- h. Analyse Repetition
- i. Analyse Alignment
- j. Analyse Proximity
- k. Analyse Aesthetic
- l. Analyse Intuitiveness

Test Results:

- g. Analyse Contrast
 - Colour contrast is present
- h. Analyse Repetition
 - Follows a similar formatting to the journal screen
- i. Analyse Alignment
 - The screen follows a centre alignment
- j. Analyse Proximity
 - Elements are spaced out evenly
- k. Analyse Aesthetic
 - Consistent adaptation of handwritten feel
- l. Analyse Intuitiveness
 - Consistent adaptation of style

Additional Comments: Performs as expected.

2.5 Analytics System

2.5.1 Test Case: Functionality Testing - Analytics

The main purpose of this test case is to check if the analytics screen performs its expected function of displaying analytics. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Viewing different journal log details
- b. Refresh journals and logs
- c. Toggling relative
- d. Analyse graphs provided
- e. Other considerations

Test Results:

- a. Viewing different journal log details
 - Each journal displays their respective logs
- b. Refresh journals and logs
 - Refreshes journal and displays the first journal in the database
 - Seems redundant as changes made to any journal or logs have already been updated once the user navigates back to the analytics screen due to recomposition
- c. Toggling relative
 - The purpose of this button is to toggle the relative axis for the graph but users are unlikely to intuitively understand this functionality
 - Performs as expected though
- d. Analyse graphs provided
 - Bar graph of water given over time does not work properly, a bar with a value of 340 would have the same height as a bar with a value of 546. This error can be seen across a range of values.
 - Line graph performs as expected
 - Pages are available once the journal exceeds 7 logs
- e. Other considerations
 - Unique journal ID is present in the name which could interrupt the user experience

Additional Comments: The Analytics display page generally works as expected but enhancements can be done to remove unnecessary elements and fix graph errors.

2.5.2 Test Case: Interface Testing - Analytics

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- m. Analyse Contrast
- n. Analyse Repetition
- o. Analyse Alignment
- p. Analyse Proximity
- q. Analyse Aesthetic
- r. Analyse Intuitiveness

Test Results:

- m. Analyse Contrast
 - Colour contrast is present
- n. Analyse Repetition
 - Follows the original green theme
- o. Analyse Alignment
 - Elements are aligned properly
 - Scrollable column height is set weirdly such that the elements cut off before touching the top and bottom navigation
- p. Analyse Proximity
 - Inconsistent spacing between certain elements with some really far apart and some touching such as the buttons.
 - Buttons displayed also have inconsistent sizing.
- q. Analyse Aesthetic
 - The layout is appealing to the eyes with easy-to-read graphs
- r. Analyse Intuitiveness
 - Button functionalities can be confusing due to the lack of explanation

Additional Comments: Generally well-designed but improvements can be made to fix errors and enhance user experiences such as a proper scrolling view.

2.6 Profile System

2.6.1 Test Case: Functionality Testing - Side Drawer (Profile Icon)

The main purpose of this test case is to check if the side drawer performs its expected function of provided profile functions. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. **Display side drawer**
- b. **Sign in when accessing from an offline view**
- c. **Display the user sign-in name**
- d. **Sign out**

Test Results:

- a. **Display side drawer**
 - Works as expected when using a physical Android device
 - Unable to access side drawer while on emulator
- b. **Sign in when accessing from an offline view**
 - Content made during sign-out offline will be linked up to your account once you turn on wifi and sign in useful feature
- c. **Display the user sign-in name**
 - Works as expected
- d. **Sign out**
 - Works as expected

Additional Comments: The side drawer generally works as expected but like many other elements in the top navigation, it cannot be accessed via the emulator due to a lack of padding considerations.

2.6.2 Test Case: Interface Testing - Side Drawer (Profile Icon)

The main purpose of this test case is to analyse the screen's design based on the 4 basic principles of User Experience / User Design, aesthetic, and intuitiveness.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- s. Analyse Contrast**
- t. Analyse Repetition**
- u. Analyse Alignment**
- v. Analyse Proximity**
- w. Analyse Aesthetic**
- x. Analyse Intuitiveness**

Test Results:

- s. Analyse Contrast**
 - Colour contrast is present
- t. Analyse Repetition**
 - Follows standard side drawer view
- u. Analyse Alignment**
 - Elements are aligned
- v. Analyse Proximity**
 - Elements are spaced out evenly
- w. Analyse Aesthetic**
 - No design was made to improve the side drawer
- x. Analyse Intuitiveness**
 - Button functionalities are intuitive

Additional Comments: Works as expected.

2.7 Offline Mode

2.7.1 Test Case: Functionality Testing - Offline Mode

The main purpose of this test case is to check if the application still functions during offline mode where applicable. Other functionalities present on the screen will also be checked.

Test Owner: Respeto Sofia Amihan Molase

Test Date: 18 February 2025

Test Steps:

- a. Turn on the internet connection, log in, and try offline mode
- b. Turn on the internet connection, log in, try offline mode, and turn off the internet connection while in the application
- c. Turn on the internet connection, log in, try offline mode, and turn off the internet connection while in the application, make changes, and then log back in
- d. Turn off the internet connection, log in, try offline mode, then turn on internet connection, log in, try offline mode
- e. Turn off the internet connection, sign in, try offline mode, then turn on internet connection, log in, try offline mode

Test Results:

- a. Turn on the internet connection, log in, and try offline mode
 - Works as expected
- b. Turn on the internet connection, log in, try offline mode, and turn off the internet connection while in the application
 - Works as expected where the remote API functionalities do not display data and the user is unable to log out of their account
- c. Turn on the internet connection, log in, try offline mode, and turn off the internet connection while in the application, make changes, and then log back in
 - Works as expected where offline changes sync up once the internet connection is back
- d. Turn off the internet connection, log in, try offline mode, then turn on internet connection, log in, try offline mode
 - Works as expected where the remote API functionalities do not display data and the user is unable to log out of their account
 - Works as expected where offline changes sync up once the internet connection is back
- e. Turn off the internet connection, sign in, try offline mode, then turn on internet connection, log in, try offline mode
 - Works as expected where offline changes sync up once the internet connection is back

Additional Comments: Works well as a caching system that can link to the account and server once the internet connection is turned back on.

3.0 Test Assessment

For the test assessment, I adopted a screen-by-screen basis where I would divide the tests by functionality and interface. For each scenario, I would examine everything present on the screen and create different test cases, especially for forms. In terms of functionality, I ensured that features performed as expected while identifying opportunities to enhance the user experience. As for the interface, I applied the C.R.A.P design principles — Contrast, Repetition, Alignment, Proximity — to determine whether the element choices were appropriate. Finally, I tested the application on both an emulator and a physical device to evaluate its responsiveness and adaptability.

4.0 Test Result

4.1 Functional Testing

The table below summarises the test cases employed for ad hoc testing and the test results obtained for each test case:

Test Case ID	Date Tested	Tester	Pass/Fail	Severity of Defect	Summary of Defect	Comments
2.1.1	18-02-2025	Sofia	Pass	-	-	-
2.1.3			Pass	-	-	-
2.2.1			Pass	-	-	-
2.2.3			Fail	Moderate	The application can crash due to the ability of journal buttons to be accessed mid-transition. However, the chances of this happening to a user are slim as no one usually does these series of taps.	-
2.2.5			Fail	Minor	It is too easy to delete journals which could lead to weeks of entry logs instantly deleted. Some users may feel frustrated due to the long-winded navigation needed to make one journal entry.	-
2.3.1			Fail	Severe	Temperature and humidity functions aren't working but it's okay because these aren't compulsory fields. However, occasionally the add logs screen freezes infinitely, interrupting the user experience severely.	-
2.3.3			Fail	Moderate	The application can crash due to the ability of entry buttons to be accessed mid-transition. However, the chances of this happening to a user are slim as no one usually does these series of taps.	-

2.3.5			Fail	Severe	The share icon does not work. Unable to post the log publicly.	
2.4.1			Pass	-	-	-
2.5.1			Fail	Severe	The screen contains various elements that are redundant, with the bar graph not working as expected. The relative functionality can be confusing to users due to a lack of explanation provided by the missing onboarding experience.	-
2.6.1			Fail	Minor	Side Drawer does not work on the emulator.	-
2.7.1			Pass	-	-	-

4.2 Interface Testing

The table below summarises the test cases employed for ad hoc testing and the test results obtained for each test case:

Test Case ID	Date Tested	Tester	Pass/Fail	Severity of Defect	Summary of Defect	Comments
2.1.2	18-0-2025	Sofia	Pass	-	-	No defect but general improvements can be made to the interface where applicable.
2.1.4			Pass	-	-	
2.2.2			Pass	-	-	
2.2.4			Pass	-	-	
2.2.6			Pass	-	-	
2.3.2			Pass	-	-	
2.3.4			Pass	-	-	
2.3.6			Fail	Moderate	Various functionalities were not aware of due to poor placement.	
2.4.2			Pass	-	-	
2.5.2			Fail	Moderate	Lack of a vertical scroll height therefore the elements cut off before reaching the top and bottom navigation. Button functionalities lack an explanation as to what they do.	
2.6.2			Fail	Moderate	Lack of padding considerations prevent the user from accessing any top navigation functionalities when using the emulator.	

5.0 Variances

Overall, the application performs as expected. However, some unexpected yet intentional functionalities were observed, such as the authentication system not being linked to content created by the same user, which was not immediately apparent during my initial impression of the app.

Certain errors were either anticipated by the application or resulted in completely unexpected behaviour, such as the 'share log to public' icon not functioning, which prevented me from testing the remote API system. While all features work properly on a physical mobile device, the top navigation elements did not when tested on an emulator. Additionally, minor issues were observed in the graphs, temperature, and humidity readings, but these issues can be patched during the next update of the application.

The most unusual issue encountered was the application freezing or crashing without any clear explanation. Because Leaf Log is still in early development, there is still room for improvement in resolving these minor configuration errors. However, as a first test, the application generally meets expectations as a platform for recording plant data.

6.0 Recommendations

- During the development process, the app could be tested on multiple devices to account for responsiveness and adaptability.
- The developer could include more user support for the different kinds of errors.